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Legendre Polynomials and Functions

91. Legendre's equation and its solution. (Agra 86, 88, 90;
Bombay 76; I.A.S. 73, 76; Meerut 84, 87, 92; Kanpur 91)

The differential equation of the form

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0 \quad \dots(1)$$

is called Legendre's equation; where n is a positive integer. We now solve (1) in series of descending powers of x .

Let the series solution of (1) be

$$y = \sum_{m=0}^{\infty} c_m x^{k-m}, c_0 \neq 0 \quad \dots(2)$$

Differentiating (2) and then putting the values of y , dy/dx and d^2y/dx^2 into (1) we have

$$(1-x^2) \sum_{m=0}^{\infty} c_m (k-m)(k-m-1) x^{k-m-2} - 2x \sum_{m=0}^{\infty} c_m (k-m) x^{k-m-1} + n(n+1) \sum_{m=0}^{\infty} c_m x^{k-m} = 0$$

$$\text{or } \sum_{m=0}^{\infty} c_m (k-m)(k-m-1) x^{k-m-2} - \sum_{m=0}^{\infty} c_m \{(k-m)(k+m-1) + 2(k+m) - n(n+1)\} x^{k-m} = 0 \quad \dots(3)$$

But the coefficient of $c_m x^{k-m}$

$$\begin{aligned} &= (k-m)^2 - (k-m) + 2(k-m) - n(n+1) \\ &= (k-m)^2 - n^2 + (k-m) - n \\ &= (k-m+n)(k-m-n) + (k-m-n) \\ &= (k-m-n)(k-m+n+1). \end{aligned}$$

Hence (3) may be re-written as

$$\sum_{m=0}^{\infty} c_m (k-m)(k-m-1) x^{k-m-2}$$

$$-\sum_{m=0}^{\infty} c_m (k-m-n) (k-m+n+1) x^{k-m} \equiv 0 \quad \dots(4)$$

(4) is an identity. To get the indicial equation, we equate to zero the coefficient of the highest power of x , namely x^k in (4) and obtain

$$c_0 (k-n) (k+n+1) = 0 \text{ or } (k-n) (k+n+1) = 0, c_0 \neq 0 \quad \dots(5)$$

So the roots of (4) are $k=n$, $-(n+1)$. They are unequal and differ by an integer. The next lower power of x is $k-1$. So equating to zero the coefficient of x^{k-1} in (4), we get

$$c_1 (k-1-n) (k+n) = 0. \quad \dots(6)$$

For $k=n$ and $-(n+1)$, neither $(k-1-n)$ nor $(k+n)$ is zero. So $c_1=0$ from (6).

Finally, equating to zero the coefficient of x^{k-m} in (4), we have

$$c_{m-2} (k-m+2) (k-m+1) - c_m (k-m-n) (k-m+n+1) = 0$$

$$\therefore c_m = \frac{(k-m+2) (k-m+1)}{(k-m-n) (k-m+n+1)} c_{m-2} \quad \dots(7)$$

Putting $m=3, 5, 7\dots$ in (7) and noting that $c_1=0$, we have

$$c_1 = c_3 = c_5 = c_7 = \dots = 0, \quad \dots(8)$$

which hold good for both $k=n$ and $k=-(n+1)$.

To obtain $c_2, c_4, c_6\dots$ etc., we consider two cases.

Case I. When $k=n$. Then (7) becomes

$$c_m = -\frac{(n-m+2) (n-m+1)}{m (2n-m+1)} c_{m-2} \quad \dots(9)$$

Putting $m=2, 4, 6\dots$ in (8), we have

$$c_2 = -\frac{n(n-1)}{2(2n-1)} c_0$$

$$c_4 = -\frac{(n-2)(n-3)}{4(2n-3)} c_2 = \frac{n(n-1)(n-2)(n-3)}{2.4.(2n-1)(2n-3)} c_0$$

and so on.

Re-writing (2), we have for $k=n$

$$y = c_0 x^n + c_1 x^{n-1} + c_2 x^{n-2} + c_3 x^{n-3} + c_4 x^{n-4} + \dots \quad \dots(10)$$

Using (8) and the above values of $c_2, c_4, c_6\dots$ etc., (10) becomes (after replacing c_0 by a)

$$y = a \left[x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2.4.(2n-1)(2n-3)} x^{n-4} - \dots \right] \quad \dots(11)$$

Case II. When $k=-(n+1)$. Then (7) becomes

$$c_m = \frac{(n+m-1) (n+m)}{m (2n+m+1)} c_{m-2} \quad \dots(12)$$

Putting $m=2, 4, 6, \dots$ in (12), we have

$$c_2 = \frac{(n+1)(n+2)}{2(2n+3)} c_0$$

$$c_4 = \frac{(n+3)(n+4)}{4(2n+5)} c_2 = \frac{(n+1)(n+2)(n+3)(n+4)}{2 \cdot 4(2n+3)(2n+5)} c_0$$

and so on. For $k=-(n+1)$, (2) gives

$$y = c_0 x^{-n-1} + c_1 x^{-n-2} + c_2 x^{-n-3} + c_3 x^{-n-4} + c_4 x^{-n-5} + \dots \quad \dots(13)$$

Using (8) and the above values of $c_2, c_4, c_6, \dots$ etc., (13) becomes (after replacing c_0 by b)

$$y = b \left[x^{-n-1} + \frac{(n+1)(n+2)}{2(2n+3)} x^{-n-3} + \frac{(n+1)(n+2)(n+3)(n+4)}{2 \cdot 4(2n+3)(2n+5)} x^{-n-5} + \dots \right] \quad \dots(14)$$

Thus two independent solutions of (1) are given by (11) and (14). If we take $a = [1.3.5 \dots (2n-1)]/n!$, the solution (11) is denoted by $P_n(x)$ and is called Legendre's function of the first kind or Legendre's polynomial of degree n . Notice that (2) is a terminating series and so it gives rise to a polynomial of degree n . Thus $P_n(x)$ is a solution of (1). Again, if we take $b = n!/[1.3 \dots (2n+1)]$ the solution (14) is denoted by $Q_n(x)$ and is called Legendre's function of the second kind. Since n is a positive integer, (14) is an infinite or non-terminating series and hence $Q_n(x)$ is not a polynomial. Thus $P_n(x)$ and $Q_n(x)$ are two linearly independent solutions of (1). Hence the general solution of (1) is

$$y = A P_n(x) + B Q_n(x), \quad \dots(15)$$

where A and B are arbitrary constants.

Remark 1. When there is no confusion regarding the variable x , we shall use a shorter notation P_n for $P_n(x)$ and P_n' for $dP_n(x)/dx$ and so on.

Remark 2. Legendre's polynomial of degree n is denoted and defined by

$$P_n(x) = \frac{1.3.5 \dots (2n-1)}{n!} \left[x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4(2n-1)(2n-3)} x^{n-4} - \dots \right] \quad \dots(1)$$

We now re-write (1) in a compact form. The general term of polynomial (1) is given by

$$\frac{1.3.5 \dots (2n-1)}{n!} \cdot (-1)^r \frac{n(n-1) \dots (n-2r+1)}{2.4 \dots 2r(2n-1)(2n-3) \dots (2n-2r+1)} \times x^{n-2r} \quad \dots(2)$$

$$\begin{aligned} \text{Now } & 1.3.5 \dots (2n-1) \\ &= \frac{1.2.3.4.5.6 \dots (2n-1).2n}{2.4.6 \dots 2n} \end{aligned}$$

(4)

$$\begin{aligned}
 &= \frac{(2n)!}{(2.1)(2.2)(2.3)\dots(2.n)} \\
 &= \frac{(2n)!}{2^n.1.2.3\dots n} \\
 &= \frac{(2n)!}{2^n.n!} \quad \dots(3)
 \end{aligned}$$

$$\begin{aligned}
 &n(n-1)\dots(n-2r+1) \\
 &= \frac{n(n-1)(n-2r+1)(n-2r)(n-2r-1)\dots 3.2.1}{(n-2r)(n-2r-1)\dots 3.2.1}
 \end{aligned}$$

$$= \frac{n!}{(n-2r)!} \quad \dots(4)$$

$$2.4.6\dots 2r = (2.1)(2.2)(2.3)\dots(2.r)$$

$$= 2^r r! \quad \dots(5)$$

$$(2n-1)(2n-3)\dots(2n-2r+1)$$

$$= \frac{(2n)(2n-1)(2n-2)(2n-3)\dots(2n-2r+2)(2n-2r+1)}{(2n)(2n-2)(2n-4)\dots(2n-2r+2)}$$

$$\times \frac{(2n-2r)!}{(2n-2r)!}$$

$$= \frac{(2n)(2n-1)(2n-2)\dots(2n-2r+1)(2n-2r)}{(2n-2r-1)\dots 3.2.1}$$

$$= \frac{(2n)!}{2^n.n(n-1)(n-2)\dots(n-r+1)(2n-2r)!}$$

$$= \frac{(2n)!}{2^n(2n-2r)!} \times \frac{(n-r)(n-r-1)\dots 3.2.1}{n(n-1)(n-2)\dots(n-r+1)(n-r)}$$

$$= \frac{(2n)!}{2^n(2n-2r)!} \times \frac{(n-r)!}{n!} \quad \dots(6)$$

Using (3), (4), (5) and (6), the general term (2) becomes

$$\frac{(2n)!}{2^n(n!)^2} (-1)^r \frac{n!}{(n-2r)!} \times \frac{1}{2^r r!} \times \frac{2^n(2n-2r)! n!}{(2n)!(n-r)!} x^{n-2r}$$

$$\text{i.e. } (-1)^r \frac{(2n-2r)!}{2^n r! (n-r)!(n-2r)!} x^{n-2r} \quad \dots(7)$$

Since (1) is polynomial of degree n , r must be chosen so that

$$n-2r \geq 0 \text{ i.e. } r \leq n/2$$

Thus, if n is even, r goes from 0 to $\frac{1}{2}n$, while if n is odd, r goes from 0 to $\frac{1}{2}(n-1)$; that is, for the complete polynomial (1), r goes from 0 to $[\frac{1}{2}n]$,

$$\text{where } [\frac{1}{2}n] = \begin{cases} \frac{1}{2}n & \text{if } n \text{ is even} \\ \frac{1}{2}(n-1) & \text{if } n \text{ is odd} \end{cases}$$

Hence the Legendre polynomial of degree n is given by

$$P_n(x) = \sum_{r=0}^{[\frac{1}{2}n]} (-1)^r \frac{(2n-2r)!}{2^n r! (n-r)! (n-2r)!} x^{n-2r} \dots (8)$$

(Agra 89 ; Kanpur 73, 74, 88)

92. Legendre's function of the first kind or Legendre's polynomial of degree n . Definition.

(Kanpur 80 ; Agra 66 ; Meerut 71, 72, 73, 74 ; Gujrat 70 ; Poona 70 ; Vikram 70)

Legendre's polynomial of degree n is denoted and defined by

$$P_n(x) = \frac{1.3.5\dots(2n-1)}{n!} \left[x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2.4.(2n-1)(2n-3)} x^{n-4} - \dots \right] \dots (1)$$

or
$$P_n(x) = \sum_{r=0}^{[\frac{1}{2}n]} (-1)^r \frac{(2n-2r)!}{2^n r! (n-r)! (n-2r)!} x^{n-2r} \dots (2)$$

where $[\frac{1}{2}n] = \begin{cases} \frac{1}{2}n & \text{if } n \text{ is even} \\ \frac{1}{2}(n-1) & \text{if } n \text{ is odd} \end{cases} \dots (3)$

Legendre's function of the second kind. Definition.

This is denoted and defined by

$$Q_n(x) = \frac{n!}{1.3\dots(2n+1)} \left[x^{-(n+1)} + \frac{(n+1)(n+2)}{2(2n+3)} x^{-(n+3)} + \frac{(n+1)(n+2)(n+3)(n+4)}{2.4.(2n+3)(2n+5)} x^{-(n+5)} + \dots \right] \dots (4)$$

Remark. Using the definition (1) or (2), the first few Legendre's polynomials are given by

$$P_0(x) = 1, P_1(x) = x, P_2(x) = \frac{1}{2}(3x^2 - 1), P_3(x) = \frac{1}{2}(5x^3 - 3x),$$

$$P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3), P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x)$$

93. Generating function for Legendre polynomials.

Theorem. $(1-2xz+z^2)^{-1/2} = \sum_{n=0}^{\infty} z^n P_n(x), |x| \leq 1, |z| < 1$

(Rohilkhand 92 ; Kanpur 86 ; Agra 74 ; Raj. 79, 83)

or To show that $P_n(x)$ is the coefficient of z^n in the expansion of $(1-2xz+z^2)^{-1/2}$ in ascending powers of z .

(Agra 72 ; Meerut 73, 79, 81, 85 Rajasthan 75, 79)

[Note : $(1-2xz+z^2)^{-1/2}$ is called the generating function for Legendre polynomial $P_n(x)$].

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Proof. Since $|z| < 1$ and $|x| \leq 1$, we have

$$\begin{aligned} & (1-2xz+z^2)^{-1/2} \\ &= [1-z(2x-z)]^{-1/2} \\ &= 1 + \frac{1}{2}z(2x-z) + \frac{1.3}{2.4} z^2 (2x-z)^2 + \dots + \frac{1.3 \dots (2n-3)}{2.4 \dots (2n-2)} z^{n-1} \\ & \times (2x-z)^{n-1} + \frac{1.3 \dots (2n-1)}{2.4 \dots (2n)} z^n (2x-z)^n + \dots \end{aligned} \quad \dots(1)$$

Now, the coefficient of z^n in $\frac{1.3 \dots (2n-1)}{2.4 \dots (2n)} z^n (2x-z)^n$

$$\begin{aligned} &= \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots (2n)} (2x)^n \\ &= \frac{1.3.5 \dots (2n-1) 2^n x^n}{(2.1)(2.2)(2.3) \dots (2.n)} = \frac{1.3.5 \dots (2n-1)}{2^n . n!} 2^n . x^n \\ &= \frac{1.3.5 \dots (2n-1)}{n!} x^n. \end{aligned} \quad \dots(2)$$

Again, the coeff. of z^n in $\frac{1.3 \dots (2n-3)}{2.4 \dots (2n-2)} z^{n-1} (2x-z)^{n-1}$

$$\begin{aligned} &= \frac{1.3 \dots (2n-3)}{(2.1)(2.2) \dots 2(n-1)} \{-(n-1)(2x)^{n-2}\} \\ &= -\frac{1.3 \dots (2n-3)}{2^{n-1} 1.2.3 \dots (n-1)} \cdot \frac{2n-1}{n} \cdot \frac{n}{2n-1} [(n-1) \times 2^{n-2} \times x^{n-2}] \\ & \quad \text{(on multiplying and dividing by } (2n-1)/n) \\ &= -\frac{1.3 \dots (2n-1)}{n!} \cdot \frac{n(n-1)}{2(2n-1)} x^{n-2} \end{aligned} \quad \dots(3)$$

and so on. Using (2), (3), ..., we see that the coefficient of z^n in the expansion of $(1-2xz+z^2)^{-1/2}$, namely (1) is given by

$$\begin{aligned} & \frac{1.3.5 \dots (2n-1)}{n!} \left[x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} \right. \\ & \quad \left. + \frac{n(n-1)(n-2)(n-3)}{2.4.(2n-1)(2n-3)} x^{n-4} - \dots \right] \end{aligned}$$

i.e. $P_n(x)$, by definition of Legendre polynomial.

We find that $P_1(x), P_2(x), \dots$ will be the coefficients of $z, z^2, \dots$ in the expansion of $(1-2xz+z^2)^{-1/2}$. Thus, we may write

$$(1-2xz+z^2)^{-1/2} = 1 + z P_1(x) + z^2 P_2(x) + \dots + z^n P_n(x) + \dots$$

$$\text{or } (1-2xz+z^2)^{-1/2} = \sum_{n=0}^{\infty} z^n P_n(x).$$

9.4. Examples (Type 1) based on Art. 9.2 and 9.3

✓ **Ex. 1.** Prove : $1 + \frac{1}{2} P_1(\cos \theta) + \frac{1}{3} P_2(\cos \theta) + \dots$
 $= \log \left[(1 + \sin \frac{1}{2} \theta) / \sin \frac{1}{2} \theta \right].$

Sol. From the generating function, we have

$$\sum_{n=0}^{\infty} z^n P_n(x) = (1 - 2xz + z^2)^{-1/2} \quad \dots(1)$$

Integrating (1) w.r.t. z from 0 to 1, we get

$$\sum_{n=0}^{\infty} \int_0^1 z^n P_n(x) dz = \int_0^1 \frac{dz}{\sqrt{1 - 2xz + z^2}} \quad \dots(2)$$

Replacing x by $\cos \theta$ on both sides, (2) gives

$$\sum_{n=0}^{\infty} P_n(\cos \theta) \int_0^1 z^n dz = \int_0^1 \frac{dz}{\sqrt{1 - 2z \cos \theta + z^2}}$$

or
$$\sum_{n=0}^{\infty} P_n(\cos \theta) \cdot \left[\frac{z^{n+1}}{n+1} \right]_0^1 = \int_0^1 \frac{dz}{\sqrt{(z - \cos \theta)^2 + \sin^2 \theta}}$$

or
$$\sum_{n=0}^{\infty} \frac{P_n(\cos \theta)}{n+1} = \left[\log \{ (z - \cos \theta) + \sqrt{(z - \cos \theta)^2 + \sin^2 \theta} \} \right]_0^1$$

$$= \log \{ (1 - \cos \theta) + \sqrt{(1 - \cos \theta)^2 + \sin^2 \theta} \} - \log (1 - \cos \theta)$$

$$= \log \{ (1 - \cos \theta) + \sqrt{2(1 - \cos \theta)} \} - \log (1 - \cos \theta)$$

$$= \log \frac{(1 - \cos \theta) + \sqrt{2} \sqrt{1 - \cos \theta}}{1 - \cos \theta}$$

$$= \log \frac{\sqrt{1 - \cos \theta} \sqrt{1 - \cos \theta} + \sqrt{2} \sqrt{1 - \cos \theta}}{\sqrt{1 - \cos \theta} \sqrt{1 - \cos \theta}}$$

$$= \log \frac{\sqrt{1 - \cos \theta} + \sqrt{2}}{\sqrt{1 - \cos \theta}}$$

$$= \log \frac{\sqrt{2 \sin^2 \frac{1}{2} \theta} + \sqrt{2}}{\sqrt{2 \sin^2 \frac{1}{2} \theta}} = \log \frac{1 + \sin \frac{1}{2} \theta}{\sin \frac{1}{2} \theta}$$

$$\therefore \frac{P_0(\cos \theta)}{1} + \frac{1}{2} P_1(\cos \theta) + \frac{1}{3} P_2(\cos \theta) + \dots = \log \frac{1 + \sin \frac{1}{2} \theta}{\sin \frac{1}{2} \theta}$$

or
$$1 + \frac{1}{2} P_1(\cos \theta) + \frac{1}{3} P_2(\cos \theta) + \dots$$

$$= \log \frac{1 + \sin \frac{1}{2} \theta}{\sin \frac{1}{2} \theta} \quad [\because P_0(\cos \theta) = 1]$$

Ex. 2. Prove that (i) $P_n(1) = 1$ (ii) $P_n(-1) = (-1)^n$

(iii) $P'_n(1) = \frac{1}{2} n(n+1)$ (iv) $P'_n(-1) = (-1)^{n-1} \frac{n(n+1)}{2}$

$$(v) P_{2n}(0) = (-1)^n \frac{(2n)!}{2^{2n} (n!)^2} \quad (vi) P_{2n+1}(0) = 0$$

$$(vii) P_n(-x) = (-1)^n P_n(x). \text{ Deduce } P_n(-1) = (-1)^n.$$

(Kanpur 83; Agra 84, 88; Meerut 84; Jiwaji 89)

Sol. The generating function formula is

$$(1-2xz+z^2)^{-1/2} = \sum_{n=0}^{\infty} z^n P_n(x), \quad |z| < 1, |x| \leq 1 \quad \dots(1)$$

Part (i) Putting $x=1$ in (1), we have

$$(1-2z+z^2)^{-1/2} = \sum_{n=0}^{\infty} z^n P_n(1) \quad \text{or} \quad (1-z)^{-1} = \sum_{n=0}^{\infty} z^n P_n(1).$$

Since $|z| < 1$, the binomial theorem can be used for expansion of $(1-z)^{-1}$. We then obtain

$$1+z+z^2+\dots+z^n+\dots = \sum_{n=0}^{\infty} z^n P_n(1). \quad \dots(2)$$

Equating the coefficient of z^n from both sides, (2) gives

$$P_n(1) = 1. \quad \checkmark$$

Part (ii). Putting $x=-1$ in (1), we have as before

$$(1+2z+z^2)^{-1/2} = \sum_{n=0}^{\infty} z^n P_n(-1) \quad \text{or} \quad (1+z)^{-1} = \sum_{n=0}^{\infty} z^n P_n(-1)$$

$$\text{or } 1-z+z^2-\dots+(-1)^n z^n+\dots = \sum_{n=0}^{\infty} z^n P_n(-1) \quad \dots(3)$$

Equation the coefficients of z^n from both sides, (3) gives

$$P_n(-1) = (-1)^n. \quad \checkmark$$

Part (iii). Since $P_n(x)$ satisfies Legendre's equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0,$$

we get

$$(1-x^2)P_n''(x) - 2xP_n'(x) + n(n+1)P_n(x) = 0 \quad \dots(4)$$

Putting $x=1$ in (4) and using $P_n(1)=1$, we get

$$0 - 2P_n'(1) + n(n+1) = 0 \quad \text{or} \quad P_n'(1) = \frac{1}{2}n(n+1). \quad \checkmark$$

Part (iv). Putting $x=-1$ in (4) and using $P_n(-1)=(-1)^n$, we get

$$0 + 2P_n'(-1) + n(n+1)(-1)^n = 0. \quad \checkmark$$

or

$$P_n'(-1) = -(-1)^n \cdot \frac{1}{2}n(n+1).$$

or

$$P_n'(-1) = (-1)^{n-1} \frac{1}{2}n(n+1)$$

$$[\because -(-1)^n = -(-1)^{n-1}(-1) = (-1)^{n-1}]$$

Part (v) and (vi). Putting $x=0$ in (1), we have

$$\sum_{n=0}^{\infty} z^n P_n(0) = (1+z^2)^{-1/2}$$

(9)

$$= \sum_{n=0}^{\infty} \frac{(-\frac{1}{2})(-\frac{1}{2}-1)\dots(-\frac{1}{2}-n+1)}{n!} (z^2)^n$$

$$\therefore \sum_{n=0}^{\infty} z^n P_n(0) = \sum_{n=0}^{\infty} (-1)^n \frac{1.3.5\dots(2n-1)}{2^n.n!} z^{2n}, \quad \dots(5)$$

Equating the coefficients of z^{2n} from both sides, (5) gives

$$\begin{aligned} P_{2n}(0) &= (-1)^n \frac{1.3.5\dots(2n-1)}{2^n.n!} \\ &= (-1)^n \frac{1.2.3.4.5.6\dots(2n-1)(2n)}{2^n.n! \cdot 2.4.6\dots 2n} \\ &= (-1)^n \frac{(2n)!}{2^n.n!} \frac{1}{(2.1)(2.2)(2.3)\dots(2.n)} \\ &= (-1)^n \frac{(2n)!}{2^n.n!} \cdot \frac{1}{2^n.n!} = (-1)^n \frac{(2n)!}{2^{2n}(n!)^2} \end{aligned}$$

Next, equating the coefficients of z^{2n+1} from both sides in (5), we have $P_{2n+1}(0)=0$ because R.H.S. of (5) consists of even powers of z alone.

Part (vii): Replacing x by $-x$ in (1), we get

$$(1+2xz+z^2)^{-1/2} = \sum_{n=0}^{\infty} z^n P_n(-x). \quad \dots(6)$$

Next, replacing z by $-z$ in (1), we get

$$(1+2xz+z^2)^{-1/2} = \sum_{n=0}^{\infty} (-z)^n P_n(x). \quad \dots(7)$$

From (6) and (7), we have

$$\sum_{n=0}^{\infty} z^n P_n(-x) = \sum_{n=0}^{\infty} (-1)^n z^n P_n(x). \quad \dots(8)$$

Equating the coefficients of z^n from both sides, (8) gives

$$P_n(-x) = (-1)^n P_n(x). \quad \dots(9)$$

Remark. When n is odd, $(-1)^n = -1$ and so (9) becomes $P_n(-x) = -P_n(x)$. Hence $P_n(x)$ is an odd function of x when n is odd. Similarly $P_n(x)$ is an even function of x when n is even.

Ex. 3. Show that $P_n(0)=0$ if n is odd, and

$$P_n(0) = (-1)^{n/2} \frac{1.3.5\dots(2n-1)}{2.4\dots n} \cdot (-1)^{\frac{n}{2}} \frac{n!}{2^{n/2}((n/2)!)^2}$$

if n is even. (Kanpur 83 ; 85 ; Meerut 84 ; Agra 67)

Sol. Proceed like parts V and VI of Ex. 2 above and obtain

$$P_{2n+1}(0)=0, P_{2n}(0)=(-1)^n \frac{1.3.5...(2n-1)}{2^n.n!}$$

From these results the required results follow.

Ex. 4. Prove that

$$P_n(-\frac{1}{2})=P_0(-\frac{1}{2})P_{2n}(\frac{1}{2})+P_1(-\frac{1}{2})P_{2n-1}(\frac{1}{2})+\dots+P_{2n}(-\frac{1}{2})P_0(\frac{1}{2}).$$

$$\text{Sol. We have } (1-2xz+z^2)^{-1/2}=\sum_{n=0}^{\infty} z^n P_n(x) \quad \dots(1)$$

Replacing x by $1/2$ and $-1/2$ successively, (1) gives

$$(1-z+z^2)^{-1/2}=\sum_{n=0}^{\infty} z^n P_n(\frac{1}{2}) \quad \dots(2)$$

and

$$(1+z+z^2)^{-1/2}=\sum_{n=0}^{\infty} z^n P_n(-\frac{1}{2}) \quad \dots(3)$$

Next, replacing z by z^2 in (3), we have

$$(1+z^2+z^4)^{-1/2}=\sum_{n=0}^{\infty} z^{2n} P_n(-\frac{1}{2}) \quad \dots(4)$$

$$\text{But } 1+z^2+z^4=(1+z^2)^2-z^2=(1+z^2+z)(1+z^2-z)$$

$$\therefore (1+z^2+z^4)^{-1/2}=(1+z+z^2)^{-1/2} \cdot (1-z+z^2)^{-1/2}$$

$$\text{or } \sum_{n=0}^{\infty} z^{2n} P_n(-\frac{1}{2})=\sum_{n=0}^{\infty} z^n P_n(-\frac{1}{2}) \cdot \sum_{n=0}^{\infty} z^n P_n(\frac{1}{2}), \text{ by (2), (3) and (4)}$$

$$\text{or } \sum_{n=0}^{\infty} z^{2n} P_n(-\frac{1}{2})=[P_0(-\frac{1}{2})+z P_1(-\frac{1}{2})+\dots+z^{2n-1} P_{2n-1}(-\frac{1}{2})$$

$$+z^{2n} P_{2n}(-\frac{1}{2})+\dots] \times [P_0(\frac{1}{2})+z P_1(\frac{1}{2})+\dots+z^{2n-1} P_{2n-1}(\frac{1}{2})+z^{2n} P_{2n}(\frac{1}{2})+\dots]$$

Equating the coefficients of z^{2n} from both sides of the above equation, we get the required result.

$$\text{Ex. 5. Prove: } \frac{1-z^2}{(1-2xz+z^2)^{3/2}}=\sum_{n=0}^{\infty} (2n+1) z^n P_n.$$

Jiw aji 82 ; Rohikhand 89 ; Agra 79 ; Raj. 91

$$\text{Sol. We have } (1-2xz+z^2)^{-1/2}=\sum_{n=0}^{\infty} z^n P_n \quad \dots(1)$$

Differentiating both sides w.r.t. ' z ', (1) gives

$$-\frac{1}{2} (1-2xz+z^2)^{-3/2} (-2x+2z)=\sum_{n=0}^{\infty} n z^{n-1} P_n$$

$$\text{or } (x-z) (1-2xz+z^2)^{-3/2}=\sum_{n=0}^{\infty} n z^{n-1} P_n \quad \dots(2)$$

Multiplying both sides of (2) by $2z$, we have

$$2z(x-z)(1-2xz+z^2)^{-3/2} = 2 \sum_{n=0}^{\infty} n z^n P_n \quad \dots(3)$$

Adding (1) and (3), we have

$$\begin{aligned} & \frac{1}{(1-2xz+z^2)^{1/2}} + \frac{2z(x-z)}{(1-2xz+z^2)^{3/2}} \\ &= \sum_{n=0}^{\infty} z^n P_n + \sum_{n=0}^{\infty} 2nz^n P_n \end{aligned}$$

or
$$\frac{1-2xz+z^2+2z(x-z)}{(1-2xz+z^2)^{3/2}} = \sum_{n=0}^{\infty} (2n+1) z^n P_n$$

or
$$\frac{1-z^2}{(1-2xz+z^2)^{3/2}} = \sum_{n=0}^{\infty} (2n+1) z^n P_n.$$

Ex. 6. Prove
$$\frac{1+z}{z\sqrt{(1-2xz+z^2)}} - \frac{1}{z} = \sum_{n=0}^{\infty} (P_n + P_{n+1}) z^n$$

(Meerut 89 ; Kanpur 77 ; Agra 83)

Sol. We have $(1-2xz+z^2)^{-1/2} = \sum_{n=0}^{\infty} z^n P_n \quad \dots(1)$

$\therefore$ L.H.S. of the required result

$$= (1/z) (1-2xz+z^2)^{-1/2} + (1-2xz+z^2)^{-1/2} - (1/z)$$

$$= (1/z) \sum_{n=0}^{\infty} z^n P_n + \sum_{n=0}^{\infty} z^n P_n - (1/z), \text{ by (1)} \quad \dots(2)$$

$$\begin{aligned} \text{But } \sum_{n=1}^{\infty} z^n P_n &= zP_0 + z^2P_1 + z^3P_2 + \dots + z^n P_n + z^{n+1} P_{n+1} + \dots \\ &= 1 + zP_1 + z^2P_2 + \dots + z^n P_n + \dots (\because P_0=1) \\ &= 1 + \sum_{n=1}^{\infty} z^n P_n \end{aligned} \quad \dots(3)$$

Using (3) in (2), we get

L.H.S. of the required result

$$= (1/z) (1 + \sum_{n=1}^{\infty} z^n P_n) + \sum_{n=0}^{\infty} z^n P_n - (1/z)$$

$$= \sum_{n=0}^{\infty} z^n P_{n+1} + \sum_{n=0}^{\infty} z^n P_n = \sum_{n=0}^{\infty} (P_n + P_{n+1}) z^n$$

= R.H.S. of the required result.

Ex. 7. Prove that $(1-2xz+z^2)^{-1/2}$ is a solution of the equation

$$z \frac{\partial^2 (zv)}{\partial z^2} + \frac{\partial}{\partial x} \left\{ (1-x^2) \frac{\partial v}{\partial x} \right\} = 0. \quad (\text{Agra 85 ; Meerut 85})$$